4 **(a)** A radio mast transmits a signal using a transmitter with output power 15.0 W. The signal is received at a research station 3.5 km away.

(i) Calculate the power received by a receiving antenna of effective area 450 cm^2.

power received = ...W [2]

(ii) State the assumption made in your calculation of part (a)(i).

.....[1]

(iii) The amplitude of the signal at a distance of 2.5 km from the transmitter is A_0. Determine, in terms of A_0, the amplitude of the signal received at the antenna at a distance of 3.5 km.

amplitude = ... A_0 [2]

(b) **(i)** State what is meant by a polarised wave.

.....

.....[1]

(ii) A vertically polarised light of intensity I_o is incident on a polariser P with its transmission axis at an angle of 40° from the vertical as shown in Fig. 4.1. The wave then passes through another polariser Q with its transmission axis at an angle of 40° from the vertical in the other direction.

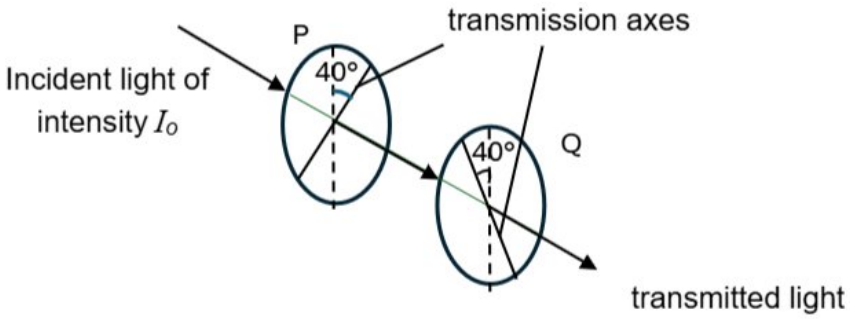


Fig. 4.1

Determine the intensity of the transmitted light in terms of I_o.

intensity = ... I_o [2]

[Total: 8]